

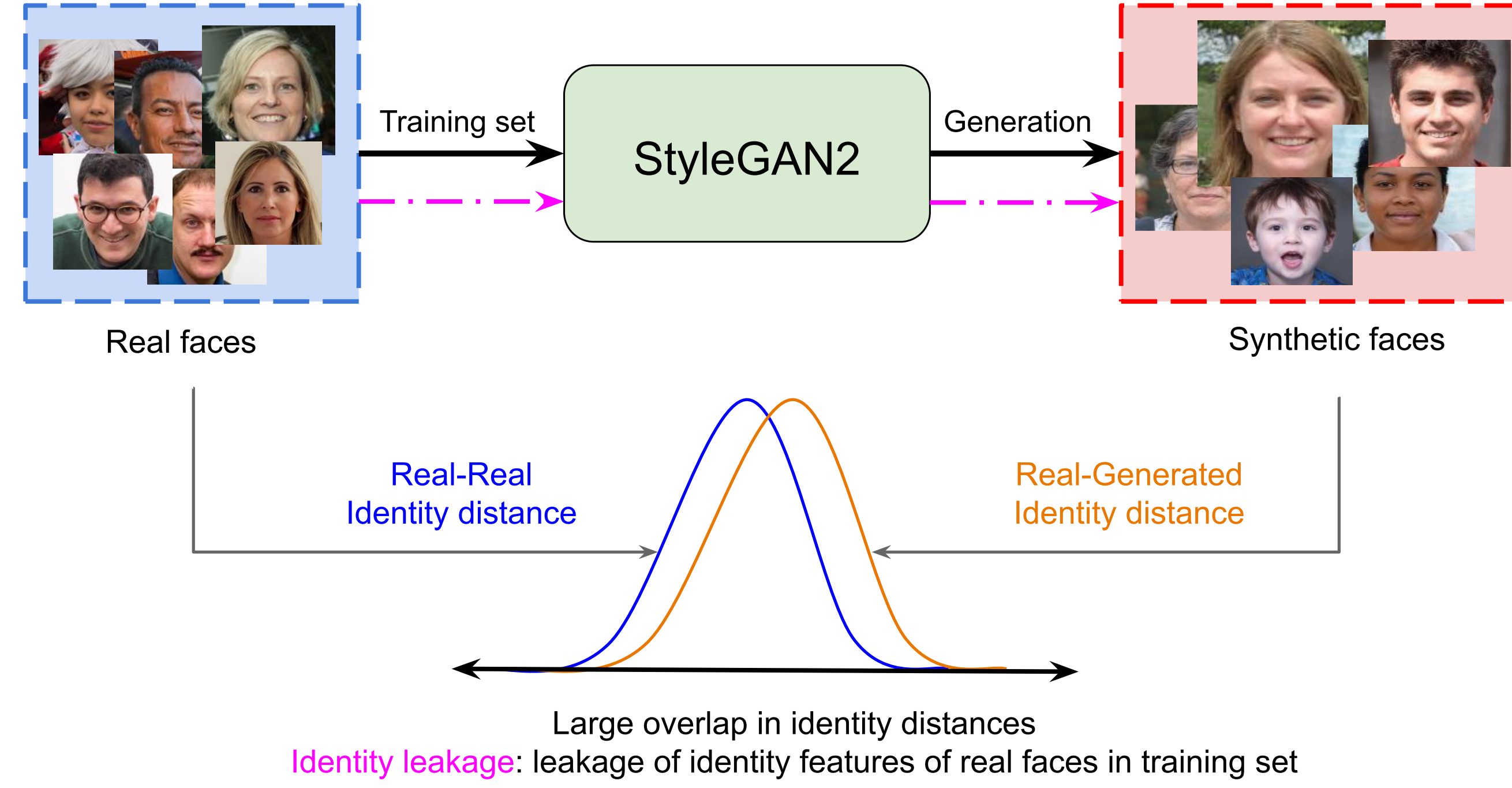
SynthProv: Interpretable Framework for Profiling Identity Leakage

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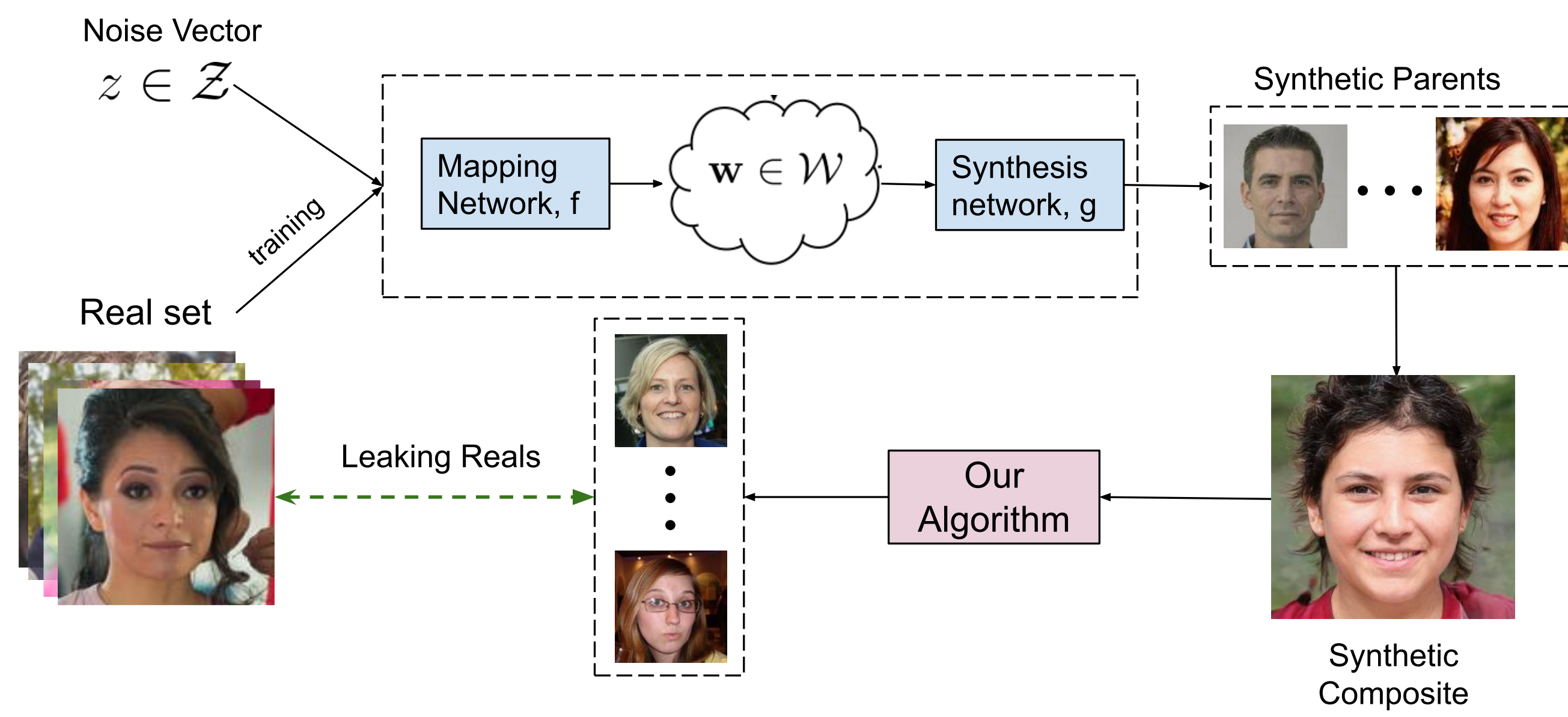
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SUMMARY

- Problem:** *Identity leakage* [1]: presence of identity features of real training faces in learnt representations of GANs.



- Aim:** *Detect and trace identity leakage* to its sources in the training set.
- Approach:** Identity leakage analysis in both
 - Synthetic image space
 - StyleGAN2 latent space
- GAN latent space represents synthetic faces as latent vectors.
- How are identity features encoded in latent representations?
- Latent space image composition to detect and trace identity leakage.

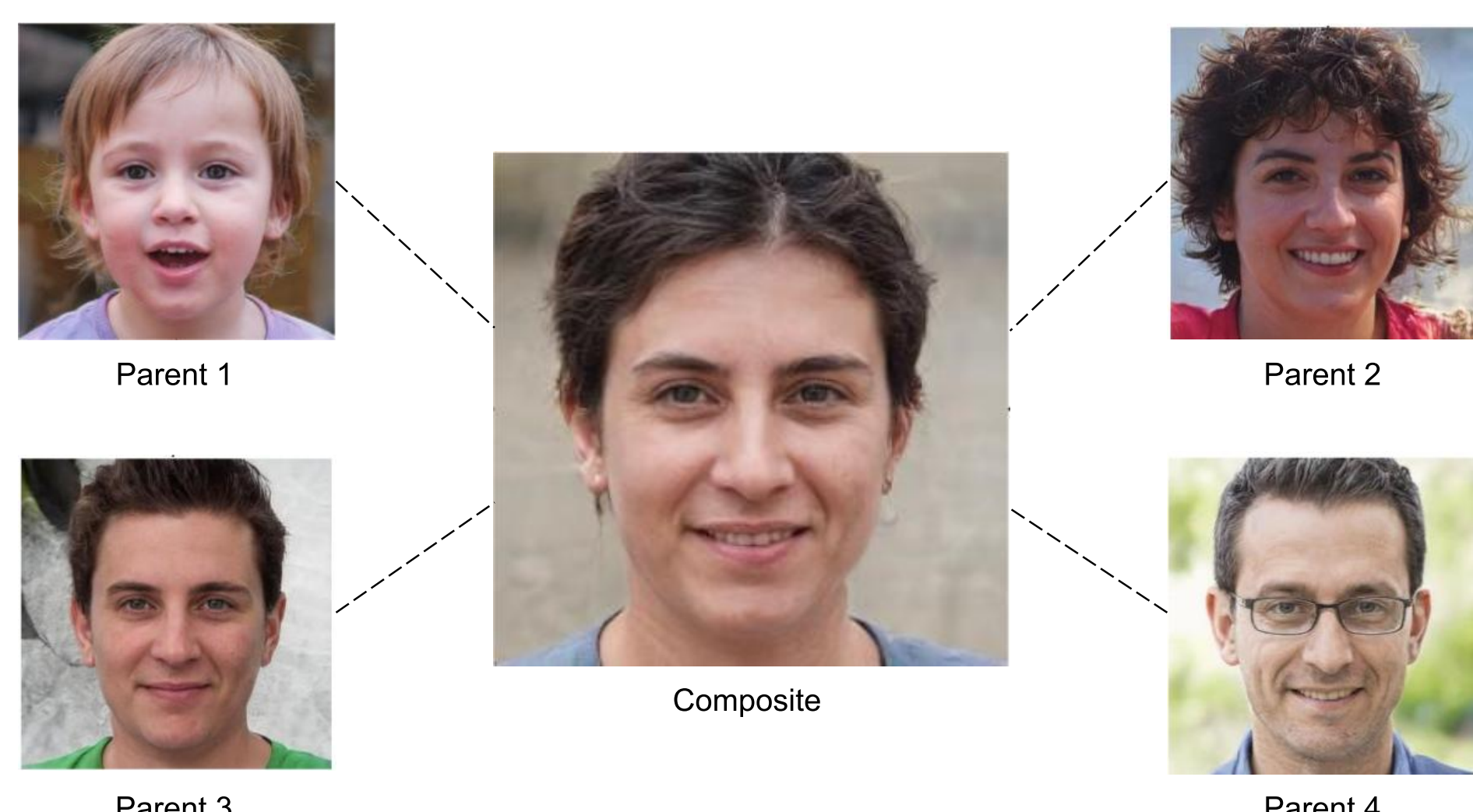


Overall framework SynthProv

IMAGE COMPOSITION

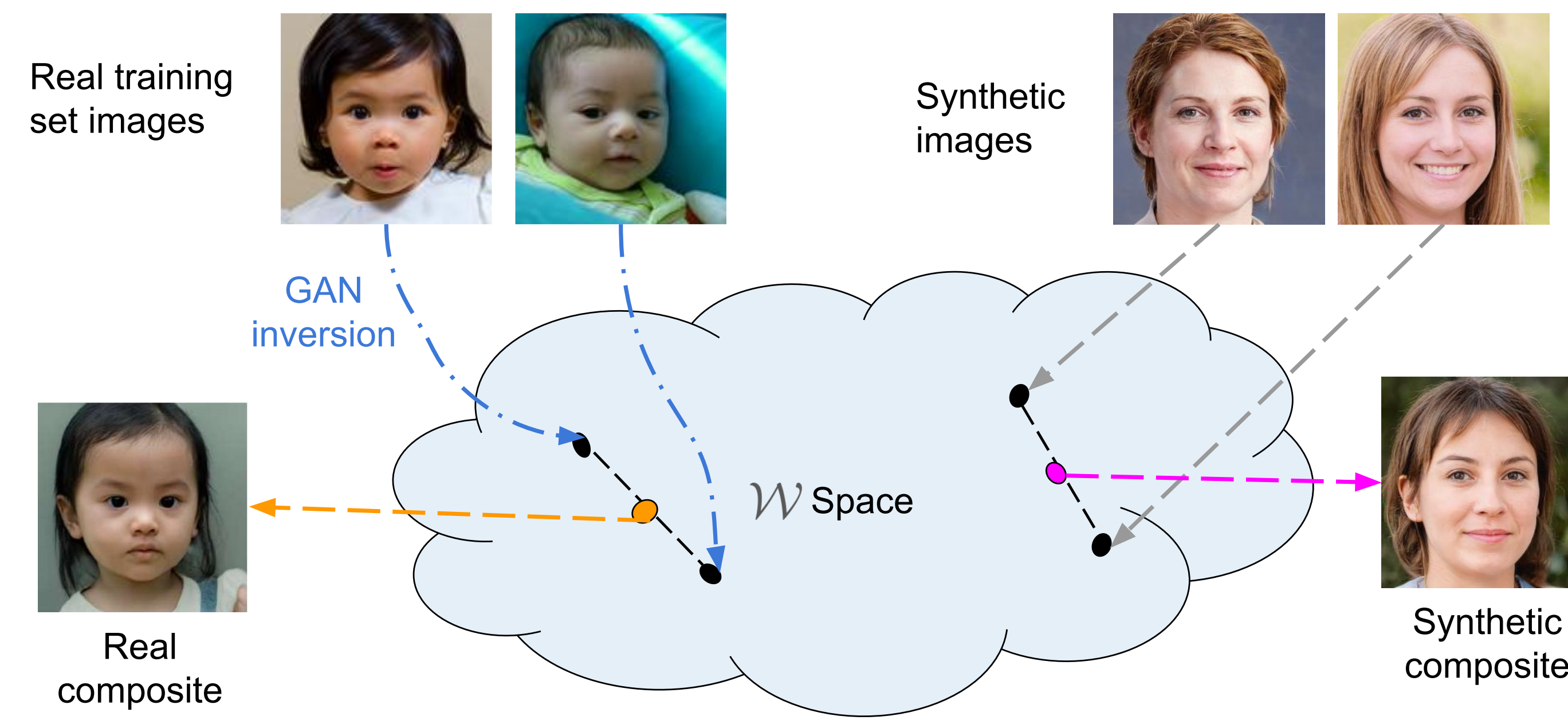
- Identities of synthetic faces are composed of real identities leaking into them.
- These identity features are encoded in the $\mathcal{W}$ space of StyleGAN2 [2].
- Composite faces are generated by averaging k parent latent vectors given by $l_{p_1}, l_{p_2}, \dots, l_{p_k} \in \mathcal{W}$ to obtain a composite latent vector l_c

$$l_c = \frac{1}{k} \sum_{i=1}^k l_{p_i}$$



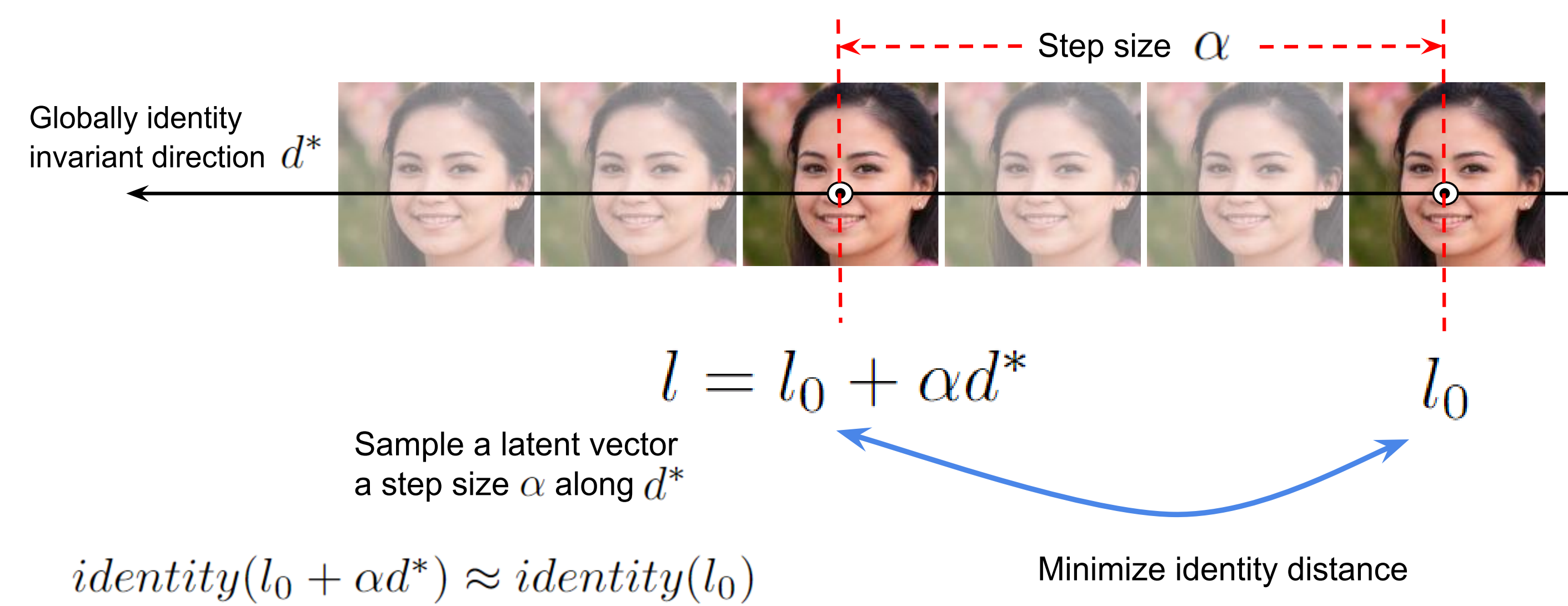
DATA GENERATION

- Two types of composites: *synthetic* and *real*
- Real composites are averages of inverted real faces from the training set given by $\mathcal{R}$.
- Synthetic composites are averages of randomly sampled latent vectors contained in the set $\mathcal{S}$.
- 10k composites of each type (synthetic and real) are generated from FFHQ and CelebAHQ for each value of $k \in \{2, 3, 4, 5, 6\}$

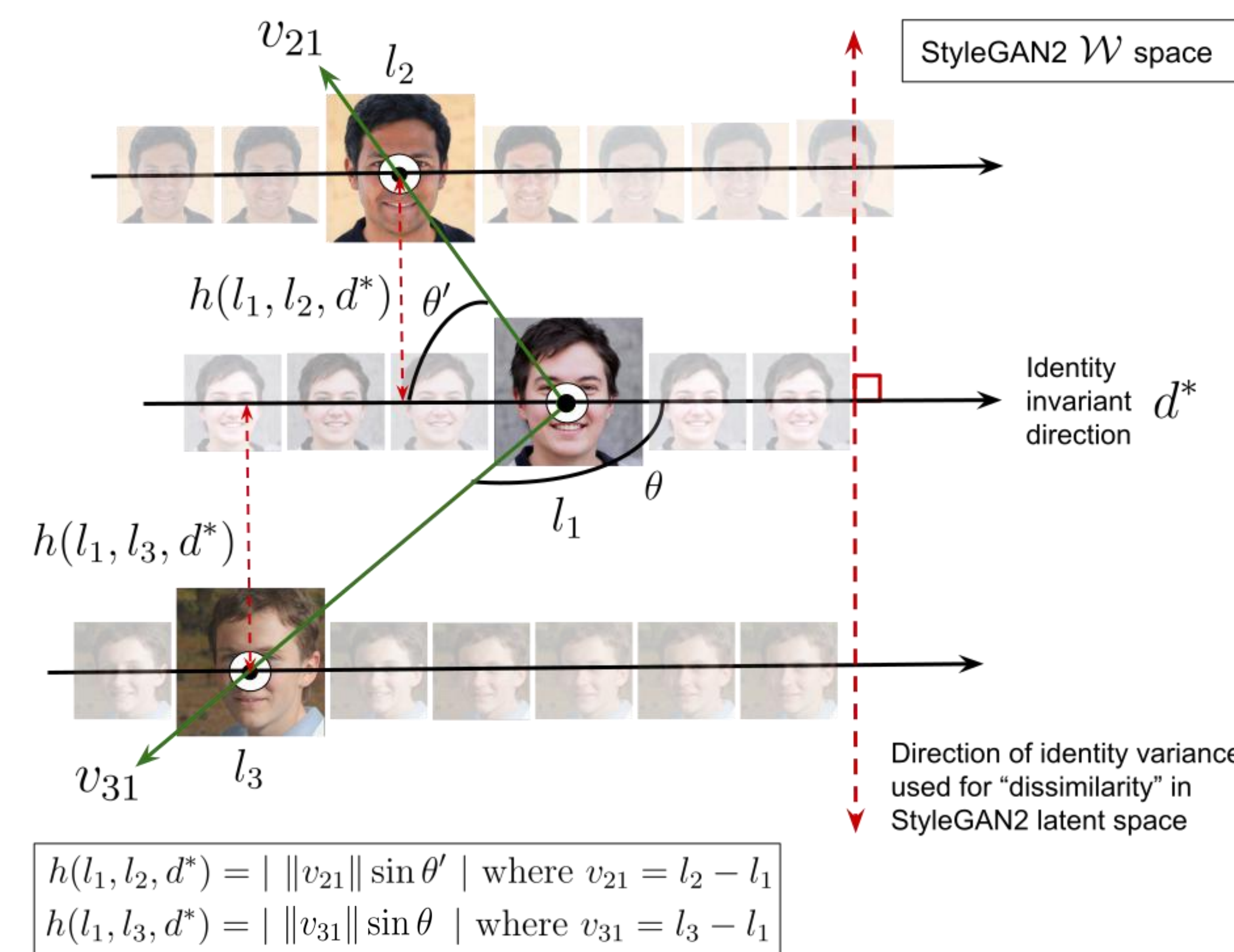


LATENT SPACE IDENTITY DISTANCE

- StyleGAN2 $\mathcal{W}$ space has semantically meaningful directions.
- A globally identity invariant direction is identified.



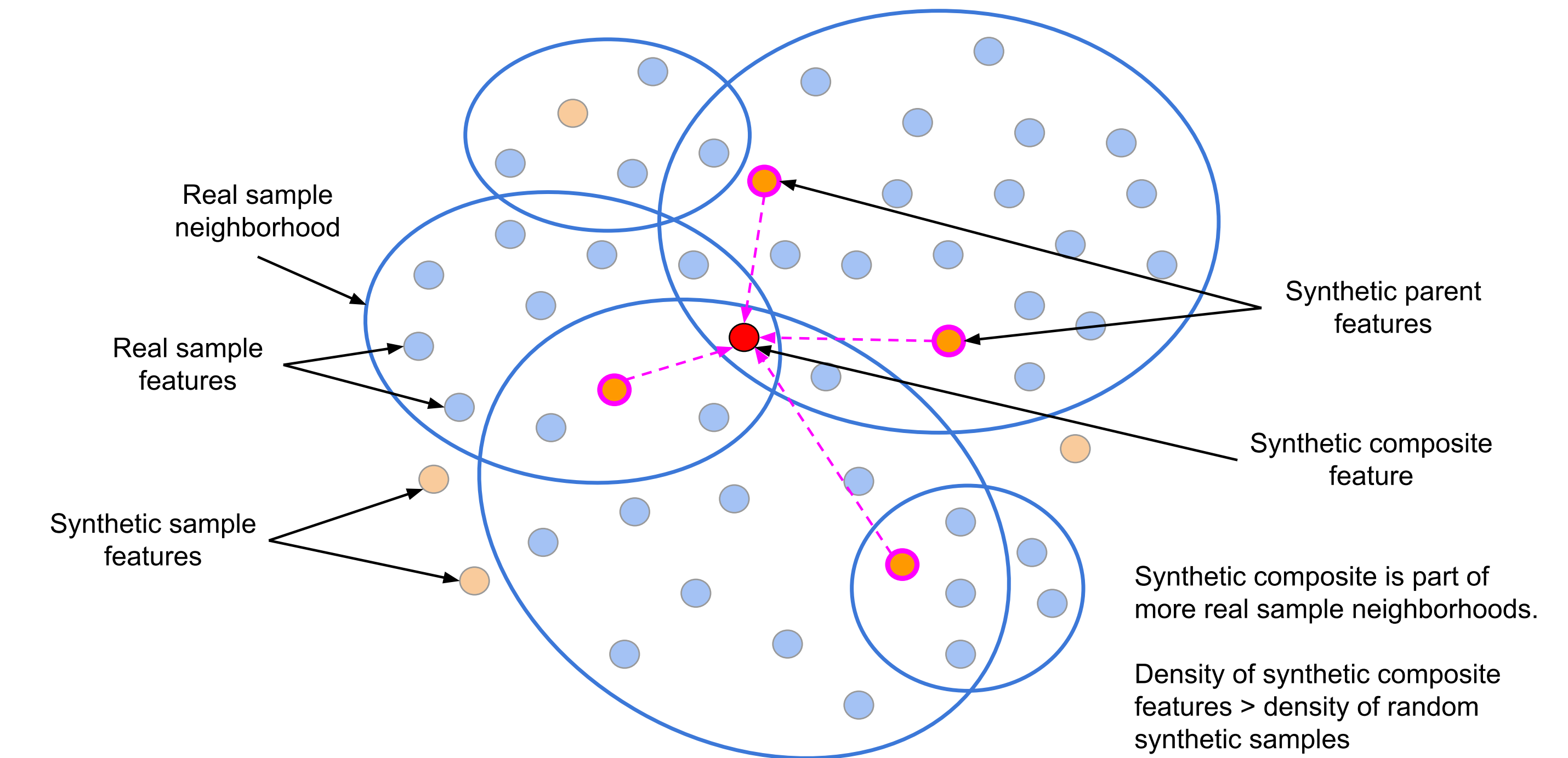
- Identity invariant direction d^* is used to find the identity dissimilarity between two latent vectors (representations of two faces).
- How do we find identity variance using d^* ?



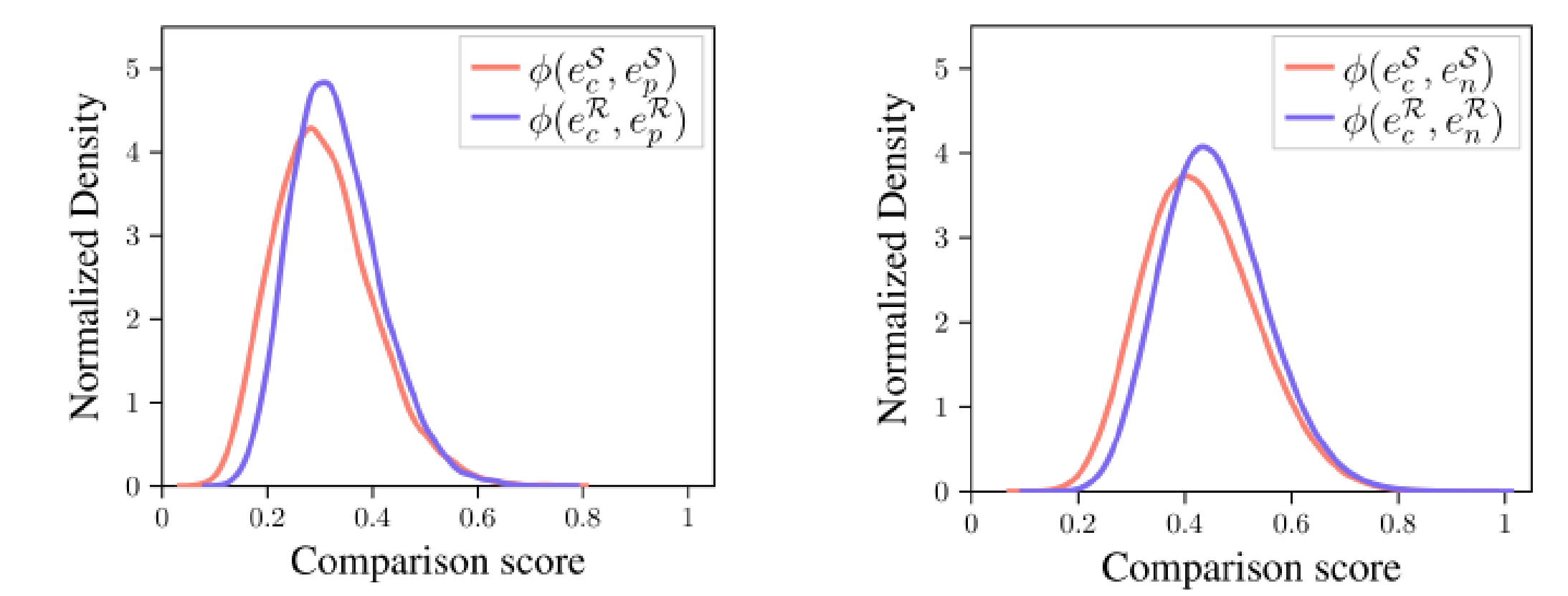
- Direction perpendicular to d^* encodes identity variance or dissimilarity.
- Projection of vector differences along perpendicular direction to gives the latent space identity distance given function $h(\cdot, \cdot, d^*)$

EXPERIMENTS & RESULTS

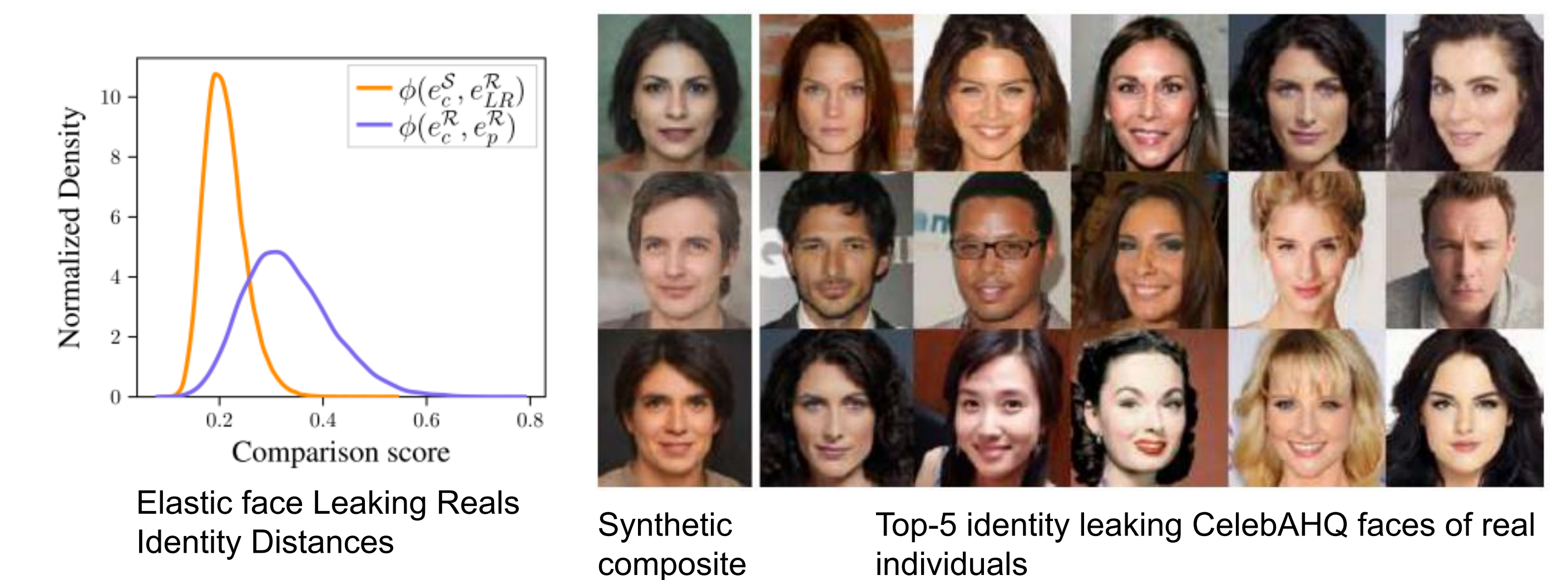
- Density metric** [5]: number of synthetic samples part of real sample neighborhoods in a feature space.
- Density in face-matcher spaces - *ElasticFace* [3] and *ArcFace* [4].
 - Shows synthetic composites touch more real samples than their parents.
 - More real identity contribution, more leakage.



- Detection of identity in the StyleGAN2 latent space and synthetic image space using composites.
 - Synthetic composites match very closely to their parents and non-parents than real composites.
 - $\Rightarrow$ identity features from parents accumulate in synthetic composites
 - $\Rightarrow$ averaged identity features remain far from unique real identities



- Tracing identity features present in synthetic composites to real faces in training set.
 - Use *proxies with shared identity features* and match to real faces in training set.
 - Obtain *matching scores* for provenance using both latent space identity distance and face-matchers.
 - Ranked matching for *Leaking Reals* for a synthetic composite which are closer to real faces than real composites.



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